UNDERSTANDING OBJECT HALLUCINATION IN VISION-LANGUAGE MODELS

Jogada Ramakanth

IIT(ISM) Dhanbad
Computer Science And Engineering
Dhanbad, Jharkhand

ABSTRACT

Vision-Language Models (VLMs) can answer questions about images but may produce incorrect predictions by claiming that an absent object is present, a behavior known as object hallucination. This study investigates binary object-existence predictions under different image conditions. Three comparatively small VLMs, referred to as **Qwen/Qwen2.5-VL-3B-Instruct**,**Qwen/Qwen3-VL-2B-Instruct**,and **HuggingFaceTB / SmolVLM2-2.2B-Instruct**, are evaluated using a dataset containing 989 unique image-question pairs with positive and negative ground-truth labels. Five image degradation methods—blur, heavy blur, resizing, compression, and occlusion—are applied to investigate their effect on **Qwen/Qwen2.5-VL-3B-Instruct**. Additional experiments compare the three models on original images, examine the influence of removing person-related(most frequently occurred questions) questions, and analyse model responses to a uniform grey image [1]. Accuracy, precision, recall, F1-score, false positive rate, and false negative rate are used to evaluate classification performance. The study provides a controlled framework for analysing how image quality, model selection, and question composition relate to object-existence predictions. It also identifies free-form caption evaluation as a direction for future work [2].

Index Terms— Vision-Language Models, Object Hallucination, Image Degradation, Multimodal Evaluation, Binary Classification.

1. INTRODUCTION

Vision-Language Models combine visual information processing with natural language understanding to perform tasks such as visual question answering, image description, and image understanding. Despite their capabilities, these models may generate answers that are not supported by the input image. One important reliability issue is object hallucination, in which a model claims that an object is present even though the ground-truth annotation indicates that it is absent.

Image degradation (blur, heavy blur, resizing, compression, and occlusion) may affect predictions, causing false positives or false negatives. Model selection and question composition may also influence evaluation outcomes. We therefore examine performance across three VLMs and assess sensitivity to question composition by removing person-related questions(most frequently occurred questions).

The study focuses on binary object-existence evaluation and does not claim to cover every form of hallucination in unrestricted image descriptions.

2. METHODOLOGY

2.1. Dataset and Models

The dataset contains 989 unique image-question pairs divided into positive and negative ground-truth subsets.Duplicate entries were removed.

Each example consists of an image, a question about the presence of an object, and a ground-truth Yes/No label. The same question is used when evaluating an original image and its corresponding degraded versions, allowing predictions across conditions to be compared consistently.

Three pretrained vision-language models were evaluated as mentioned in abstract.

2.2. Image Degradation Methods

Five image degradation methods are evaluated:
Blur: Uses a 15×15 kernel to smooth the image and reduce edge, texture, and fine-detail visibility.
Heavy blur:Uses a 51×51 kernel, causing stronger smoothing that can obscure fine details and larger object structures.
Resizing:Reduces the image to 10% of its original dimensions and enlarges it back, potentially losing fine visual information.
Compression:Uses JPEG compression at quality 5, which can introduce visual artifacts and reduce image detail.
Occlusion:Hides the central 25% of the image area, limiting the visual evidence available to the model.

Each degradation condition is compared with the original-image baseline using the same questions. The experiment uses one implemented setting for each transformation; therefore, it does not represent a complete evaluation across all possible degradation severities.

2.3. Experimental Design

The study comprises four experiments. All experiments use accuracy, precision, recall, F1-score, false positive rate (FPR), and false negative rate (FNR) for evaluation, where applicable.

Image Degradation: Qwen2.5-VL-3B is evaluated on original images and images subjected to five degradation methods, using positive and negative examples to assess changes in performance and error patterns.

Model Comparison: Qwen2.5-VL-3B, Qwen3-VL-2B, and SmolVLM2-2.2B are evaluated on original images using the same questions to compare their performance.

Person-Question Removal: The question "Is there a person in the image?" is removed, and the original-image results for all three models are compared before and after removal.

Uniform Grey Image: All three models receive a uniform grey image with questions from both positive and negative subsets. Their Yes/No responses are examined to assess behaviour without recognizable scene content; this is a diagnostic experiment, as the original ground-truth labels do not describe the grey image.

3. EVALUATION METRICS

The first three experiments use six classification metrics. A positive prediction corresponds to Yes, and a negative prediction corresponds to No.

True Positive (TP): Ground truth is Yes and prediction is Yes.

True Negative (TN): Ground truth is No and prediction is No.

False Positive (FP): Ground truth is No but prediction is Yes. This represents an incorrect positive object-presence claim.

False Negative (FN): Ground truth is Yes but prediction is No. This represents a missed positive example.

The evaluation metrics are calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5}$$

$$\text{FNR} = \frac{FN}{FN + TP} \tag{6}$$

In binary object-existence evaluation, FPR corresponds to the hallucination rate when hallucination is defined as predicting an absent object as present. However, FPR is not equivalent to every possible definition of hallucination rate.

Table 1. Performance Under Image Degradation

Condition	Acc.	Prec.	Rec.	F1	FPR	FNR
Original	0.877	0.848	0.914	0.880	0.160	0.086
Blur	0.808	0.800	0.814	0.807	0.198	0.186
Heavy Blur	0.705	0.721	0.656	0.687	0.248	0.344
Occlusion	0.823	0.844	0.787	0.815	0.143	0.213
Compression	0.780	0.756	0.818	0.786	0.258	0.182
Resizing	0.712	0.713	0.699	0.706	0.276	0.301

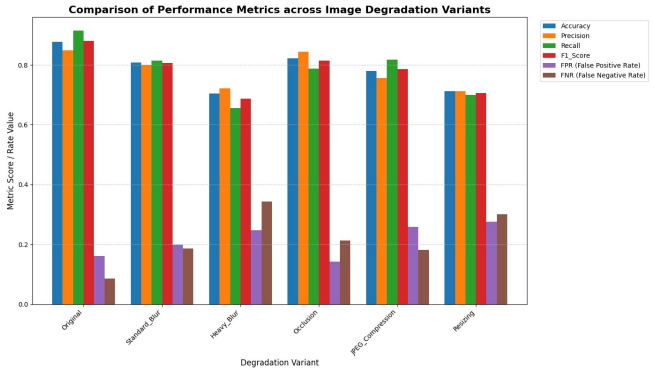


Fig. 1. Performance metrics of Qwen2.5-VL-3B across image degradation variants.

4. RESULTS AND ANALYSIS

The results section will report the measured outcomes from the four experiments. The following subsections provide the structure for inserting the actual plots and tables.

4.1. Effect of Image Degradation

Qwen2.5-VL-3B-Instruct was evaluated on original images and five degraded variants using the same binary object-existence evaluation procedure.

The original images produced the highest accuracy (0.877) and F1-score (0.880). Heavy blur caused the largest decrease in recall, from 0.914 to 0.656, while the FNR increased from 0.086 to 0.344. This indicates that the model missed a larger proportion of objects that were present.Resizing produced an accuracy of 0.712 and the highest FPR (0.276), indicating an increased proportion of false-positive predictions among negative examples. JPEG compression also increased the FPR to 0.258, although recall remained at 0.818.Occlusion produced comparatively better results than the other degraded conditions, with an accuracy of 0.823 and an F1-score of 0.815. It also produced the lowest FPR (0.143) across the evaluated conditions. These results show that degradation does not affect all error types uniformly: heavy blur was associated with more false negatives, whereas resizing and JPEG compression produced relatively high false-positive rates as shown in

fig [1].

4.2. Comparison Across Models

The three models were evaluated on the original images using binary object-existence questions.

Table 2. Performance Comparison Across Models

Model	Acc.	Prec.	Rec.	F1	FPR	FNR
Qwen2.5-VL-3B	0.877	0.848	0.914	0.880	0.160	0.086
Qwen3-VL-2B	0.870	0.815	0.953	0.878	0.212	0.047
SmolVLM2-2.2B	0.829	0.793	0.885	0.837	0.226	0.115

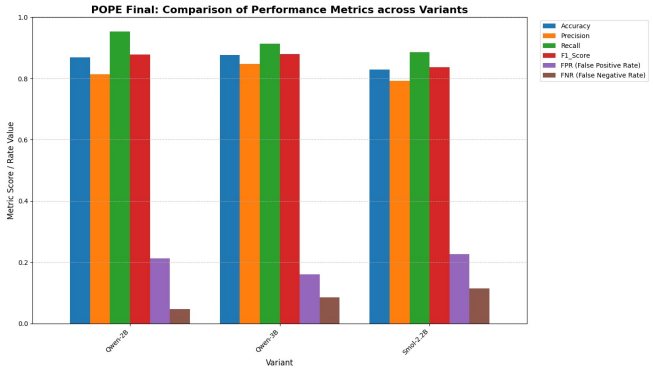


Fig. 2. Comparison of POPE evaluation metrics across the three VLMs.

Qwen2.5-VL-3B achieved the highest accuracy, precision and F1-score, as well as the lowest FPR. Qwen3-VL-2B achieved the highest recall (0.953) and lowest FNR (0.047), but its FPR was higher at 0.212. This indicates a trade-off between identifying present objects and avoiding false-positive predictions.

SmolVLM2-2.2B recorded the lowest accuracy and F1-score, together with the highest FPR. Overall, the results demonstrate that model performance depends on the metric considered: Qwen2.5-VL-3B provided the strongest overall balance, whereas Qwen3-VL-2B favoured recall. These comparisons do not establish that parameter count alone explains the differences.

4.3. Effect of Removing Person-Related Questions

To examine sensitivity to question composition, questions asking whether a person was present were removed, and the metrics were recalculated.

Table 3. Performance With and Without Person-Related Questions

Model	Condition	Acc.	Prec.	Recall	F1	FPR	FNR
Qwen3-VL-2B	With Person	0.870	0.815	0.953	0.878	0.211	0.047
Qwen3-VL-2B	Without Person	0.849	0.737	0.946	0.828	0.212	0.054
Qwen2.5-VL-3B	With Person	0.877	0.848	0.914	0.880	0.159	0.086
Qwen2.5-VL-3B	Without Person	0.851	0.773	0.869	0.819	0.160	0.131
SmolVLM2-2.2B	With Person	0.829	0.793	0.885	0.837	0.226	0.115
SmolVLM2-2.2B	Without Person	0.810	0.706	0.866	0.778	0.225	0.134

Removing person-related questions reduced accuracy, precision and F1-score for all three models. SmolVLM2-2.2B showed the largest F1-score decrease, from 0.837 to 0.778. Qwen2.5-VL-3B decreased from 0.880 to 0.819, while Qwen3-VL-2B decreased from 0.878 to 0.828.Precision declined more than recall for each model. For example, Qwen3-VL-2B's precision decreased from 0.815 to 0.737, while its recall changed from 0.953 to 0.946. This suggests that the measured false-positive behaviour was sensitive to question composition as shown in fig 3. However, because removing questions changes the evaluation subset, the comparison does not establish a causal relationship between person-related questions and model behaviour.

4.4. Responses to a Uniform Grey Image

A uniform grey image was evaluated using the same set of 989 questions. The ground-truth labels were manually reassigned to No, reflecting the absence of the queried objects in the grey images. All three evaluated systems produced No for all 989 questions.

Table 4. Model Performance on Uniform Grey Images

Model	Questions	Yes	No	Accuracy	Precision	Recall	F1
Qwen2.5-VL-3B	989	0	989	1.000	N/A	N/A	N/A
Qwen3-VL-2B	989	0	989	1.000	N/A	N/A	N/A
SmolVLM2-2.2B	989	0	989	1.000	N/A	N/A	N/A

The uniform response pattern indicates that no positive object-presence predictions occurred in this experiment. Since every ground-truth label was No, the accuracy and error rates reflect performance on an all-negative test set. These results describe model behaviour under this specific diagnostic condition; they do not demonstrate that the models never hallucinate on ordinary images.

Note:A small number of unparseable Yes/No outputs were excluded, leaving 986–989 valid rows per experiment (¡0.4% of the 989-question dataset). Metrics therefore reflect valid responses only, with invalid outputs treated as neither correct nor incorrect.

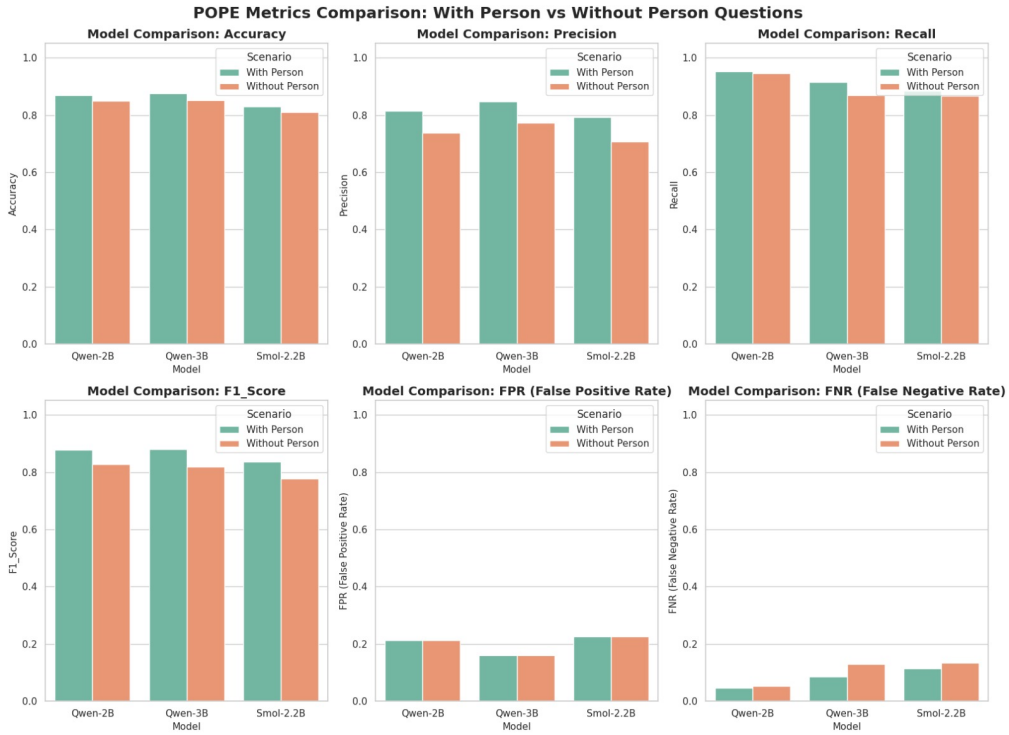


Fig. 3. Comparison of evaluation metrics with and without person-related questions.

5. CONCLUSION

This study evaluated object-existence predictions from three pretrained vision-language models under original-image, degradation, question-ablation and uniform grey-image conditions. Qwen2.5-VL-3B achieved the highest accuracy and F1-score on original images, while Qwen3-VL-2B achieved the highest recall. For Qwen2.5-VL-3B, heavy blur caused the largest recall reduction, whereas resizing and JPEG compression produced relatively high false-positive rates. Removing person-related questions reduced F1-score across all three models, demonstrating sensitivity to question composition.

In the grey-image experiment, all models answered No to all questions after the ground-truth labels were manually reassigned to No. This establishes consistent negative responding under the tested condition, but not general immunity to hallucination.

Rohrbach et al. introduced CHAIR to measure object hallucination in image captions, while Li et al. proposed POPE, which uses binary object-existence questions [1], [2]. Building on POPE, this study evaluates three pretrained VLMs on original images, examines five image degradation methods using Qwen2.5-VL-3B, and investigates the effects of removing person-related questions and using uniform grey images.

6. DISCUSSION AND LIMITATIONS

Image degradation affected object-existence predictions differently: heavy blur increased false negatives, whereas resizing and JPEG compression increased false-positive rates. Qwen2.5-VL-3B achieved the highest accuracy and F1-score, while Qwen3-VL-2B achieved the highest recall. Removing person-related questions reduced F1-scores across all models. All models predicted No for the uniform grey image, achieving perfect accuracy on the manually relabelled all-negative set; however, this does not establish general resistance to hallucination. There are several limitations to this study. Degradation was tested only on Qwen2.5-VL-3B, with one configuration per method. Binary questions do not capture hallucinations in free-form captions, while removing person-related questions changes the evaluation subset, limiting causal interpretation. Finally, a small number of invalid outputs were excluded, so the reported metrics reflect only valid responses.

7. FUTURE WORK

Future work should evaluate all three models under identical degradation conditions, investigate multiple degradation severities, and extend the evaluation to free-form captions using metrics such as CHAIR [1]. Larger and more diverse datasets could further establish whether these patterns generalize beyond the current experimental setting.

8. REFERENCES

[1] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen, "Evaluating object hallucination in large vision-language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Singapore, Dec. 2023, pp. 292–305, Association for Computational Linguistics.

[2] Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko, "Object hallucination in image captioning," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Brussels, Belgium, oct – nov 2018, pp. 4035–4045, Association for Computational Linguistics.